

TECHNICAL WHITE PAPER

Building a Publication-Ready Data Foundation for the TIN200 Report

A Week-by-Week Account of the Validation, Research and Contact Work Delivered for New Zealand's Technology Investment Network — with a Supporting Automation and Analytics Layer

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1. Executive Summary

The Technology Investment Network, or TIN, publishes the TIN200 each year. It ranks as the country's largest technology exporters, and its readership is wide: investors weigh it, government agencies quote it, and the listed firms measure themselves against it. All that influence rests on one thing, which is whether the numbers underneath are correct. That was the brief for this capstone. The task was to take the raw data behind the 2026 report and bring it to a standard where any single figure could withstand challenge.

This paper presents the project as a sequence of weekly tasks. Across eight weeks the team built the 2026 candidate list, developed the early-stage and Maverick shortlists, researched cleantech and export-economy data, verified executive contacts across the list, validated the acquisitions that alter company ownership, and assembled long-term research assets covering India–China expansion and the cleantech ecosystem. All five members shared this client's work. In addition, three members contributed a supporting technical layer; the contribution documented in detail here is an automation pipeline that cleans the raw files in approximately thirty seconds, together with an analytics layer that converts the cleaned data into business insight.

The outcomes are measurable. Thirty merger-and-acquisition events were validated against primary regulatory records, establishing that a quarter of the TIN200 is now foreign-owned. The list was confirmed to have generated NZ\$19.99 billion in revenue during 2025, approximately four times the growth rate of the wider economy, with the ten largest companies accounting for roughly half of that total. Principal component analysis demonstrated that 85.7 percent of the variation across six core financial metrics reduces to a single underlying factor, interpretable as company size. A correlation analysis confirmed that these six metrics are so closely linked that they behave as near-duplicates of one another.

The central recommendation is straightforward. The validated dataset, the four-tier sourcing method developed during the project, and the reproducible cleaning pipeline together provide TIN with a production process that can be trusted and repeated. One decision remains with the client. With a quarter of the list now foreign-owned, TIN must establish a clear policy on whether those companies remain eligible for inclusion. The project has identified and quantified the issue; the ruling itself rests with the client.

2. Introduction

2.1 Context and the business problem

For twenty years the TIN200 report has been the reference point for New Zealand's technology sector. What holds that reputation together is not glamorous work. It is clean, checkable data. One wrong acquirer, an out-of-date chief executive, or a firm that has slipped offshore without anyone noticing — any of these chips away at trust in the publication. For the 2026 edition, TIN needed a team to repair and reinforce that base, and to leave behind research that the report could keep drawing on in later years.

The scope comprised two distinct layers, and distinguishing between them is important for interpreting this paper. The core scope was the weekly client work set by TIN's research lead: building and maintaining the candidate list, developing the report shortlists, researching supporting data, verifying contacts, validating acquisitions, and producing forward-looking research. This constitutes the project itself. Beyond that core, three team members each added a technical stream extending past the client's stated requirements. That additional work is supporting value rather than the engagement itself, and the distinction is maintained throughout.

2.2 Approach and individual focus

Although the project was a team effort, this paper concentrates on the components led by the author: the Week 7 acquisition validation, directed from research design through to the final sourced dataset, and the data-cleaning automation that supports the team's workflow. A business-insight layer connecting the validated figures to the wider New Zealand economy is also documented. The work of other team members is summarised for completeness, in keeping with the requirement for a technical overview of the whole project, while the analytical depth remains concentrated on the author's own contribution.

2.3 How this paper is organised

The remainder of the paper is laid out as follows. Section 3 covers how the project was run, who did what, and the week-by-week progression of the work. Section 4 turns to the data itself and the engineering needed to make it usable. The technical core is Section 5, which works through the validation method, the M&A taxonomy, the exploratory analysis, and the supporting automation and analytics, with each clearly marked. Recommendations follow in Section 6 and the conclusion in Section 7, ahead of the references and appendices.

3. Project Overview and Methodology Framework

3.1 How the work was structured

The project ran without a formal data-mining framework, and that was a deliberate choice. What shaped it instead were the client's weekly briefs. Every week brought one defined deliverable — a candidate list, a set of contacts, an acquisition validation — and every deliverable moved through the same few steps: read the

brief, gather the data, check it against sources that could be trusted, and review the result before it went out. Responsibility for that last review sat with whoever held the rotating Group Leader role that week. The honest description is an iterative, client-led workflow rather than a textbook method. For data heading into a published report, it kept attention to where it mattered, on getting the facts right and proving them.

3.2 The team — shared core, individual streams

Every core client task was split evenly across the five members, and the Group Leader role moved from person to person each week so that responsibility for the final check never sat permanently with one individual. On top of that shared base, three members each carried a separate technical stream, listed below.

Member	Core client tasks (shared equally)	Additional technical stream
Gibson G	List, shortlists, research, contacts, M&A, EDA	Automation pipeline + business-insight analytics
Leo Chung	List, shortlists, research, contacts, M&A, EDA	Emerging-companies research and contact identification
Sai Krishna	List, shortlists, research, contacts, M&A, EDA	NZ economy six-year study + OneDrive cloud data hosting
Sreenath Suman	List, shortlists, research, contacts, M&A, EDA	Power BI dashboard; Python cleaning; OneDrive file alignment for clean Power BI connection
Xinyang Yan	List, shortlists, research, contacts, M&A, EDA	Cleantech-companies research and contact identification

3.3 The project, week by week

The work unfolded as a run of weekly client tasks, each one leaning on what came before. The table below sets out the whole progression ahead of the detailed sections. Taking in sequence the reasoning is plain enough: decide who belongs on the list, build the list out, check it, defend it against what the acquisition record shows, and then turn to assets that will serve the following year.

Week	Focus	What the team did	Key output
1-2	Build the 2026 candidate list	Reviewed last year's TIN200, flagged companies that no longer qualify (acquired, inactive, offshore), and researched new entrants using NZGCP investment data and sponsor recommendations.	Validated TIN200 working list + contact list for new entrants
3	Directory updates & shortlists	Maintained the working list and built the report's 'people's choice' shortlists — Early-Stage Companies (BNZ Top 10) and the Maverick shortlist — plus launch-speaker research and directory cleanups.	Early-Stage & Maverick shortlists; updated directory
4	Research & data development	Started the cleantech directory (per page 32 of the 2025 Report), built 2025 early-stage investment data from NZGCP Young Company Finance figures, researched 2025 export revenue for dairy, tourism, meat and forestry, summarised the CPO event transcript, and chased four	Cleantech list; early-stage & export datasets; CPO insights

		bounced contacts (Oritain, FrameCAD, Futura Trails, Argenta).	
5	New entrants, Mavericks, speaker letter	Sourced contacts for sponsor-recommended entrants, expanded the Maverick shortlist (established, Windcave-calibre firms — not startups), prepared the follow-up process, and styled Greg's Andrew Forrest speaker-invitation letter to an executive standard.	Expanded shortlist; polished speaker letter
6	Contact research & verification	Took the 83 TIN200 companies needing new contacts, split them across the five of us, and sourced direct CEO / CFO / Finance Manager contacts — verifying each before adding it, and flagging where only a generic inbox existed.	Verified contact dataset; generic-only companies flagged
7	Acquisition validation, directory, contacts	Validated acquisitions for nine named companies plus the team's earlier shortlist, reviewed the directory for CEO and ownership changes, and verified contacts with LinkedIn links. (My lead — see Section 5.2.)	30 validated M&A events; directory & contact updates
8	Expansion, media monitoring, cleantech	Built three long-term research assets: an India & China expansion database, a TIN200 media-monitoring database, and a full NZ cleantech ecosystem map (universe, company analysis, investment, investors, global trends).	Three future-Report research databases

A couple of points run through the entire sequence. The first is that data quality never behaved like a single tidy step. Corrections kept flowing back into the dataset week after week, which is precisely why the repeating weekly rhythm fitted the work so well. The second is that the nature of the task changed as the weeks went on. The early stretch, Weeks 1 through 6, was about assembling and checking the list; Week 7 was about holding it up against the messy reality of acquisitions; and Week 8 looked ahead, building material for future editions. The sections that follow dig into the places where the real analytical depth sits.

3.4 Individual focus

Within this body of work, three components are examined in detail: the Week 7 acquisition validation (Section 5.2), the automation pipeline (Section 5.4), and the business-insight layer (Section 5.5). The first is core client work led by the author; the latter two represent the additional technical value contributed on top of it.

4. Data Sources and Data Engineering

4.1 Sources and structure

The project drew on three primary spreadsheets, each one provided by TIN and each one checked further by the team against public records. Calling them a database would overstate things. They were working Excel files, everyone carrying its own quirks, and wrestling those quirks into shape took up a fair slice of the engineering time.

Source file	Role	Shape	Notes
2025_TIN200_RANKINGS.xlsx	Authoritative rank order	177 rows	Current and prior-year ranks; the file the Report ships from
TIN200_2025_Data_Staging.xlsx	Working master	PrePub 236 rows; Directory 200 rows	Financials in one sheet, descriptive fields in another
GLOBAL_DATA_2025.xlsx	Wide analytic table	200 rows x 269 cols	Five-year history plus current and prior year for every metric; exports by six regions

The unit of analysis is one company per row. The fields extend to revenue and employee counts for the current and prior year, wages, research-and-development spend, sales-and-marketing spend, EBITDA, exports across six destination regions, employment by New Zealand region, sector and ownership classifications, and ranking history. All monetary values are expressed in New Zealand dollar thousands. The Global Data file is the most detailed, with 269 columns, and it carries both 2025 and 2024 values, which allowed year-on-year analysis to be conducted from a single source.

4.2 Data quality assessment

A full audit turned up four findings worth setting down. On the surface each looks trivial. Left alone, though, every one of them would have fed a wrong number somewhere downstream.

- **Completeness.** The core financial fields are essentially complete for 2025. Just one missing R&D value across two hundred companies. The value was left missing rather than imputed, because inventing a number to fill one cell adds risk without adding information.
- **Distribution shape.** Every money variable is heavily right-skewed — revenue skewness sits above three. That single fact dictated a lot. It means the median, not the mean, is the honest measure of the typical company, and any future model needs a log transform before it behaves.
- **Outliers.** A 1.5-times-IQR test flags ninety-three outliers across revenue, employees, EBITDA and R&D. But they cluster in about ten genuine giants — Xero, Fisher & Paykel Healthcare, Fisher & Paykel Appliances. All were retained as genuine signals. Removing them would understate the sector by roughly a third. In a benchmark of the biggest companies, the biggest companies are the point.
- **A real defect.** The sector field held two spellings of the same category — one with a capital T, seventy-nine companies under one spelling and one under the other. A pivot table reads that as two sectors and splits the totals. The inconsistency was identified and normalised at the cleaning stage so it can never resurface.

4.3 Tools

The tooling was deliberately conventional. Python performed the core processing — pandas and numpy for the data and openpyxl to read and write Excel without disturbing the source formatting. A Windows batch

file wraps the Python so that a non-programmer can run the entire process with a single double-click. Power BI sits at the end of the chain for the dashboard. Full version details are provided in Appendix E.

5. Analytical Methods and Implementation

This section forms the technical core of the paper. The first three sub-sections address the core client work shared across the team — validation, M&A intelligence, and exploratory analysis. The final two address the additional value contributed on top. Each sub-section is labelled so that the boundary between the project and the supporting work remains explicit.

5.1 Core client work — the validation method

Determining whether a stated fact about a company is true required a defined system, because business media frequently contradict one another and a published report cannot tolerate error. A four-tier source hierarchy was therefore established and applied to every claim.

Tier	Source type	Examples	How we treat it
1	Regulatory / primary	NZ Takeovers Panel, NZX, OIO, Companies Office, High Court	Authoritative — accept
2	Company official	Press releases, investor decks	Trusted on the company's own facts
3	Major business media	NZ Herald, RNZ, Reuters, BusinessWire	Context and triangulation
4	Aggregators	PitchBook, Tracxn, Crunchbase	Cross-reference only, never sole source

The governing rule is straightforward: where two sources disagree, the higher tier prevails; where the higher tier is silent, the conflict is flagged, both versions are recorded, and judgement is deferred. The SmartPay case illustrates the value of this discipline. Early in 2025, business media reported a takeover bid from Tyro Payments, which was subsequently cancelled before Shift4, a United States firm, agreed terms and completed the acquisition in November. Recording 'Tyro' based on the initial reporting would have entered a falsehood into the dataset. The hierarchy required confirmation from the NZX scheme filing before anything was recorded, which is the method functioning as intended.

5.2 Core client work — M&A intelligence and the deal taxonomy

This stream was led by the author. The Week 7 brief named nine companies for review — SmartPay, Helius Therapeutics, 4RF, Fronde, illion Digital Tech Solutions, SnapComms, EasyCrypto, SwipedOn, and Frogparking — alongside the team's earlier shortlist. All nine were validated, after which the wider TIN200 was examined for further deals, identifying seventeen additional events. Thirty were recorded in total, each with acquirer, date, acquirer location, a New Zealand-based flag, deal value, source, and a conflict note.

Three structurally distinct deal types emerged, and the taxonomy was designed to handle each:

- Clean foreign acquisition. One buyer, one date. Robotics Plus to Yamaha Motor of Japan, February 2025. This type is recorded without complication.
- Two-step ownership chains. illion went to Experian, then Experian sold it to Credit Clear. SwipedOn went from SmartSpace to Sign In App. The rule adopted was to record the current owner and flag the ownership chain.
- Partial acquisition. Tiny Ltd of Canada took 66 percent of Serato while the founders kept the rest. Whether that constitutes foreign ownership is unclear. That is a policy call, not a data call, and it was therefore flagged for TIN rather than pretending to answer it.

One correction merit particular attention, as it demonstrates why primary sources outweigh reputation. Endace, an Auckland network-monitoring firm, is widely listed as foreign-owned following its acquisition by Emulex in 2013. However, the company's management repurchased it in 2016, and it is wholly New Zealand-owned today. Secondary sources never reflected the change, whereas the primary record did, and the entry was corrected accordingly.

Considered as a whole, the pattern is notable. Of the thirty deals, approximately eleven involved United States buyers, four Australian, two Japanese, two Canadian, two British, one South Korean, and one Chinese. Five years earlier the same list would have been almost entirely American and Australian. The buyer pool has become genuinely global. Most of these transactions retained the New Zealand operations, employment, and intellectual property in place — Robotics Plus remains in Tauranga and Seequent in Christchurch — which reframes the narrative from one of hollowing-out to one of global validation, a more constructive interpretation for the report to adopt.

The method underlying the sweep is worth noting, as it is reusable. Each company on the TIN200 list was cross-checked against three streams simultaneously: the NZX delisting record, since a delisting frequently signals an acquisition; the Overseas Investment Office consent register, which logs large foreign purchases requiring government approval; and targeted news searches keyed on the company name together with terms such as 'acquired', 'merger', and 'takeover'. Where a deal surfaced, it was traced to the primary filing before being recorded. This triangulation is the reason the seventeen additional deals constitute evidence rather than rumour.

5.3 Core client work — exploratory data analysis

The cleaned and validated data supported a twenty-one-tab exploratory analysis. The bulk of this analysis was descriptive — the groundwork required to understand the dataset before any advanced method is applied. The headline characteristics are as follows: total revenue of NZ\$19.99 billion across the 200 companies, a median company revenue of NZ\$29.2 million, and a strongly right-skewed distribution in which a small number of large firms sit far above a long tail of mid-sized companies. Revenue divides 49 percent to High-tech Manufacturing, 45 percent to ICT, and 6 percent to Biotech, and median export intensity stands at 70 percent. These descriptive breakdowns — by sector, region, ownership, and revenue band — constitute

most of the exploratory analysis and provide the figures on which the report relies. Two analytical methods then examined the data more deeply.

Correlation structure

Correlation comes first because it sets up everything after it. A Pearson correlation matrix was run across the six core financial variables. The strongest links are with revenue: 0.96 against employee count and 0.94 against wages, both sitting comfortably past the 0.9 mark where multicollinearity starts to cause real trouble. Revenue against EBITDA lands at 0.80. R&D and sales-and-marketing spend move together at roughly 0.80 as well, yet hold a loose tie to revenue, which suggests discretionary spending follows a slightly different logic from sheer operating scale. The upshot is simple enough: most of these measures are near copies of one another, because a larger company tends to be larger on all of them at the same time.

Principal component analysis

The correlations suggest that the six variables overlap substantially; principal component analysis confirms this and quantifies it. All six variables were standardised to zero mean and unit variance — a necessary step, since revenue expressed in millions would otherwise dominate employee counts expressed in hundreds, and the method would simply recover the largest raw figure. The data were then decomposed into principal components. The result is pronounced: the first component alone explains 85.7 percent of the total variance across the six metrics, and the first two together explain 92.1 percent. (ref: Appendix F) In substantive terms, the financial profile of a TIN200 company is largely reduced to a single dimension, interpretable as company size. The component loadings confirm this, with revenue, employees, wages, research and development, marketing, and EBITDA all loading on it almost equally. The practical consequence is direct: any future predictive model should avoid using all six variables as separate inputs, since they are near-identical and would introduce severe collinearity. One or two components capture almost all the relevant information.

5.4 Additional value — the automation pipeline

The first of the two additional contributions is described here. This work extends beyond the client's stated requirements, and that framing is stated explicitly before the description.

The underlying problem is operational. The report is refreshed annually from the same three irregular Excel files. Manual cleaning introduces two risks: it is slow, and a subsequent analyst may clean the data slightly differently, which quietly compromises year-on-year comparability. To address this, a two-layer pipeline was built, with a lightweight Windows batch file serving as the trigger and a Python script serving as the engine.

Design decisions and alternatives

Two decisions required careful consideration. The first was the choice of a batch file over PowerShell. PowerShell is the more capable tool, but Windows blocks unsigned PowerShell scripts by default, and most corporate laptops enforce that restriction; a batch file runs without such obstruction. Capability was therefore traded for the assurance that the process runs on any machine without IT involvement. The second decision was the use of deterministic rules rather than machine learning for the cleaning itself. A model could have been used to impute the single missing R&D value, but this was deliberately avoided. A

published report is audited material, every input row must yield the same output on every run, and a model introduces variation by design. Sophistication was therefore traded for reproducibility, which in this context is the overriding requirement.

Implementation

The batch script resolves its own folder location, searches for a Python interpreter in order of preference, installs the two required libraries if they are absent, runs the engine, and halts with a clear message if any step fails. The Python engine reads the three files, normalises column names, coerces financial fields to numeric values (with invalid entries becoming missing rather than halting the run), applies the sector-casing correction, computes year-on-year growth, writes three cleaned files, and logs the run with a timestamp and row count. The source files are never altered. Repeated runs produce byte-identical output, which was confirmed through a hash comparison.

Result

End to end, the process takes approximately thirty seconds and requires a single action to start. A verified run is recorded in the log dated 19 May 2026. The dashboard reads directly from the cleaned outputs, so the entire chain from raw file to visualisation is connected and repeatable. A task that previously required a careful manual session is now completed in the time it takes to fetch a coffee.

5.5 Additional value — business insight from the numbers

A clean dataset acquires value only when it yields an interpretable account. That account is the second contribution documented here. The findings below are those most relevant to the client and to any reader considering the direction of the New Zealand economy.

Concentration is the first theme. The TIN200 generated NZ\$19.99 billion in revenue during 2025, of which the ten largest companies accounted for 50.9 percent. Six companies with revenue above half a billion dollars carry 42.4 percent between them, and removing the top five reduces the entire list by roughly a third. This is a top-heavy market, and it carries a dual implication: it is a strength, in that New Zealand possesses genuine global-scale champions, and a risk, in that a setback at Xero or Fisher & Paykel Healthcare would move the national figures on its own.

The sector picture is the second theme and contains a notable feature. High-tech Manufacturing is the largest sector by revenue at 49 percent despite having fewer companies than ICT, a consequence of hardware scale. ICT, however, reinvests considerably more in research, at a median of 24 percent of revenue against 11 percent for manufacturing. The largest sector and the most research-intensive sector are therefore not the same, a distinction of relevance to any decision about the allocation of research-and-development support.

The third theme connects the data to the wider economy. The TIN200 grew at approximately nine percent annually over five years while the broader New Zealand economy grew at around two percent, indicating that the technology sector is not merely participating in the economy but driving a disproportionate share of its growth. These companies export 70 percent of their revenue at the median, against a national average

near 30 percent. The foreign-acquisition pattern is also clearer when the macroeconomic context is added: a weak New Zealand dollar through 2024 and 2025 made domestic technology firms cheaper for overseas buyers, while falling interest rates reduced the cost of financing acquisitions. The wave is therefore partly cyclical rather than purely a vote of confidence, and when the dollar recovers the pace may be expected to ease — a qualification worth presenting to report readers before the trend is over-interpreted.

A talent dimension also emerged, drawn from an industry workforce event referenced during the project. New Zealand technology firms increasingly struggle to recruit research-and-development engineers domestically and rely on immigration pathways and overseas-student channels to address the shortfall. Employers are also shifting their hiring criteria, moving away from fixed technical skills toward a combination of technical ability, interpersonal skills, and cultural fit. For a sector growing at four times the national rate, the binding constraint increasingly appears to be people rather than capital, and this is a structural risk on which the report can comment with authority.

6. Recommendations and Future Work

6.1 Technical recommendations

- Principal components should be used in place of the six raw financial variables in any predictive model, as the variables are nearly collinear and would destabilise the coefficients.
- Revenue should be log-transformed before any modelling, since the skew is severe and ordinary regression assumptions fail without it.
- Write the four-tier sourcing hierarchy into TIN's published methodology. It is the audit trail that lets the client defend any entry under challenge.
- The pipeline should be scheduled. Wrapping it in Windows Task Scheduler would convert the single-action run into an unattended quarterly refresh.
- The output should be versioned. Date-stamped output folders would provide a year-on-year audit trail directly from the pipeline.

6.2 Future technical enhancements

Three extensions would follow naturally from this foundation. Company-level revenue forecasting on the five-year panel, using log-linear regression with sector fixed effects, would allow TIN to project individual company trajectories rather than only the aggregate. Drawing live macroeconomic indicators from the Stats NZ and Reserve Bank open APIs would allow the dashboard to display the technology sector against the wider economy in real time. A natural-language classifier could automatically tag the more than one thousand emerging companies that currently lack a sector label, which represents the most significant manual bottleneck in the pipeline and the natural successor to the Week 8 research assets.

6.3 Scalability and production

By design, the current build stays small: three files in, three files out, no database, nothing in the cloud. For a team of this size working at this scale, that is the right call. It would not survive a shift to quarterly publishing, an emerging-companies dataset ten times larger, or several analysts running it at once. If any of those arrive, the sensible next step is a packaged Python application with a command-line interface, a versioned data store, and a test suite that re-runs the pipeline against a fixed reference set whenever the code changes. None of that is needed yet. For what the project requires, the lighter build is the correct one.

6.4 The business value, in one line

Stated plainly, the pipeline reduces a manual session to thirty seconds and removes the risk of a single analyst retaining the cleaning logic informally; the sourcing method provides TIN with an audit trail that can be defended publicly; and the insight layer converts a spreadsheet into the form of commentary the report's investor and government readers require. The combined effect is faster production, lower risk, and a richer account.

7. Conclusion

The objective was to render the data underlining the TIN200 trustworthy and to build research assets the report can rely on for future years. Both objectives were achieved. Week by week, the team built the 2026 candidate list, developed the early-stage and Maverick shortlists, researched cleantech and export data, verified contacts across 83 companies, validated thirty acquisitions, and produced the India–China and cleantech research databases. Each task was quality-checked before it reached the client.

The author's contribution lay in three areas. The acquisition validation was led by the author, including the detailed sweep that identified seventeen deals beyond the brief. The automation pipeline that makes the cleaning reproducible was designed and built by the author. The insight layer connecting the figures to the wider economy — concentration, the foreign-acquisition pattern, the export profile, and the emerging talent constraint — was also the author's work. None of this replaces the core client work; it adds to it.

Judging on its own technical terms, the project succeeded. Every figure is traceable to a source, the pipeline is deterministic and logged, and the analysis is reproducible by any analyst holding the three input files and this paper. The single outstanding item is not the team's to resolve: TIN must determine how it treats the quarter of its list that is now foreign-owned. The project identified the question, quantified it, and presented it with supporting evidence. That is the proper function of a technical foundation — to make the subsequent decision easier for those responsible for making it.

8. References

References follow APA 7th edition. Sources are grouped for readability.

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9. Appendices

Appendix A — Pipeline code (key excerpts)

Excerpt A.1 — Python-detection cascade and error handling in GENERATE_CLEAN_FILES.bat (Section 5.4):

```
@echo off
setlocal
set FOLDER=%~dp0
where py >nul 2>&1 && (set "PYCMD=py") || (
  where python >nul 2>&1 && (set "PYCMD=python") || (
    where python3 >nul 2>&1 && (set "PYCMD=python3") || (
      echo ERROR: Python not found. Install Python 3.10+ and retry.
      pause & exit /b 1)))
%PYCMD% -m pip install --quiet pandas openpyxl numpy
%PYCMD% "%FOLDER%\tin200_generate.py" "%FOLDER%"
if errorlevel 1 (echo PIPELINE FAILED - see log. & pause & exit /b 1)
echo SUCCESS. & pause
```

Excerpt A.2 — sector-casing fix and protected growth calculation in tin200_generate.py (Sections 4.2, 5.4):

```
# Sector-casing defect fix (Section 4.2)
df['Primary Sector'] = df['Primary Sector'].replace(
    {'High-Tech Manufacturing': 'High-tech Manufacturing'})

# Year-on-year growth, guarded against divide-by-zero and NaN
df['YoY_Growth_Pct'] = np.where(
    df['CY-1 Revenue'] > 0,
    (df['CY Revenue'] - df['CY-1 Revenue']) / df['CY-1 Revenue'] * 100,
    np.nan)
```

Appendix B — Weekly deliverable index

Week	Deliverable produced
1-2	2026 TIN200 working list (validated + new entrants); contact list
3	Early-Stage (BNZ Top 10) and Maverick shortlists; launch-speaker list; directory updates
4	Cleantech directory; 2025 early-stage investment dataset (NZGCP YCF); 2025 export-revenue dataset; CPO event summary; four bounced-contact recoveries
5	Sponsor-entrant contacts; expanded Maverick shortlist; styled Andrew Forrest speaker letter
6	Verified contact dataset for the 83 companies needing new contacts; generic-only companies flagged
7	30 validated M&A events; directory review; verified contacts with LinkedIn links
8	India & China expansion database; TIN200 media-monitoring database; NZ cleantech ecosystem map (Parts A-E)

Appendix C — Top 10 TIN200 companies by 2025 revenue

Rank	Company	Sector	Revenue (NZ\$M)
1	Xero	ICT	2,103
2	Fisher & Paykel Healthcare	High-tech Mfg	2,021
3	Datacom	ICT	1,338
4	Fisher & Paykel Appliances	High-tech Mfg	1,277
5	Wētā FX	ICT	585
6	Vista Group	ICT	570
7	Rocket Lab	High-tech Mfg	521
8	Pushpay	ICT	447
9	Seequent	ICT	430
10	Cin7	ICT	410

Appendix C.2 — Acquirer geography across 30 validated deals

Acquirer country	Deals	Representative examples
United States	~11	Shift4, Bentley, Aviat, Everbridge, KKR, Sixth Street, Apple
Australia	~4	Seisma, Swyftx, Credit Clear, Five V
Japan	2	Yamaha (Robotics Plus), Sysmex
Canada	2	Lightspeed (Vend), Tiny (Serato 66%)
United Kingdom	2	Access Group (Tradify), Sign In App (SwipedOn)
South Korea	1	Lunit (Volpara Health)
China	1	Tencent (Grinding Gear Games)

Appendix D — Data dictionary (key fields)

Field	Type	Definition	Unit
CY Revenue	numeric	Current-year total revenue	NZ\$ thousands
CY-1 Revenue	numeric	Prior-year total revenue	NZ\$ thousands
CY EmpTTL	integer	Total employees, current year	count
CY R&D	numeric	R&D expenditure, current year	NZ\$ thousands

CY EBITDA	numeric	EBITDA, current year	NZ\$ thousands
CY Total Exports	numeric	Total offshore revenue	NZ\$ thousands
Primary Sector	categorical	ICT / High-tech Manufacturing / Biotech	-
Ownership	categorical	Private / Foreign Owned / Public / Private-VC	-
CY TIN200 Rank	integer	Position in current list	1-200

Appendix E — Technical specifications

Component	Specification
Operating system	Windows 10/11 for the batch trigger; Python core is platform-agnostic
Python	3.10 or later
pandas	2.x
numpy	1.26+
openpyxl	3.x
scipy	1.11+ (KS and Welch t-tests)
Runtime	~30 seconds end-to-end on a standard laptop
Output verification	SHA-256 byte-identity confirmed across repeat runs

```

TIN200 IC5 Generate Clean FI x + v
2026-06-08 20:26:39,594 Staff_Geography : 200 rows
2026-06-08 20:26:39,594 Export_Totals_by_Region : 15 rows
2026-06-08 20:26:39,595
2026-06-08 20:26:39,595 FILE 3 - TIN_Rankings.xlsx
2026-06-08 20:26:39,595 Reading Rankings_2025 (TIN200 RANKINGS 2025)...
2026-06-08 20:26:39,604 [✓] Rankings_2025: 200 rows
2026-06-08 20:26:39,604 Reading Early_Stage_Investment...
2026-06-08 20:26:39,605 [✓] Early_Stage_Investment: source empty - kept existing data (16 rows)
2026-06-08 20:26:39,625 [✓] Early_Stage_Investment: 16 rows
2026-06-08 20:26:39,625 Building Revenue_Forecast_2026 (from rankings + sector CAGR)...
2026-06-08 20:26:39,626 [✓] Revenue_Forecast_2026: 236 rows
2026-06-08 20:26:39,663 Saving TIN_Rankings.xlsx...
2026-06-08 20:26:39,678 [✓] TIN_Rankings.xlsx saved (40 KB)
2026-06-08 20:26:39,678 Rankings_2025 : 200 rows
2026-06-08 20:26:39,679 Early_Stage_Investment : 16 rows
2026-06-08 20:26:39,679 Revenue_Forecast_2026 : 236 rows
2026-06-08 20:26:39,679
2026-06-08 20:26:39,679 ALL DONE - Power BI files ready
2026-06-08 20:26:39,680 Finished: 2026-06-08 20:26:39
2026-06-08 20:26:39,680
2026-06-08 20:26:39,680 TIN_Companies.xlsx [✓]
2026-06-08 20:26:39,680 TIN_Global.xlsx [✓]
2026-06-08 20:26:39,680 TIN_Rankings.xlsx [✓]
2026-06-08 20:26:39,680
2026-06-08 20:26:39,680 Open Power BI and click Refresh - done.
=====
SUCCESS! All 3 files generated.
=====
TIN_Companies.xlsx - Company_Directory, Company_Financials,
TIN_Global.xlsx - Export_Revenue, Staff_Geography,
TIN_Rankings.xlsx - Export_Totals_by_Region,
Rankings_2025, Early_Stage_Investment,
Revenue_Forecast_2026
Now open Power BI and click Refresh.
Press any key to continue . . . |

```


Appendix F – EDA Principal component analysis

```
# Step 1 - standardise
X_std = (X - X.mean()) / X.std()

# Step 2 - eigenvalues of the correlation matrix (largest first)
eigvals = np.sort(np.linalg.eigvalsh(np.corrcoef(X_std.T)))[::-1]

# Step 3 - proportion of variance explained by each component
ev = eigvals / eigvals.sum()

print("PRINCIPAL COMPONENT ANALYSIS")
for i, e in enumerate(ev, start=1):
    print(f"PC{i}: {e*100:5.1f}%")

print(f"\n PC1 alone      : {ev[0]*100:.1f}%")
print(f"PC1 + PC2 together: {(ev[0]+ev[1])*100:.1f}%")
```

[4] ✓ 0.0s Python

```
... PRINCIPAL COMPONENT ANALYSIS
PC1: 85.7%
PC2: 6.4%
PC3: 4.6%
PC4: 2.6%
PC5: 0.4%
PC6: 0.2%

PC1 alone      : 85.7%
PC1 + PC2 together: 92.1%
```